

Full-Hessian heat representation and regular future-insertion rational-current closure

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1 Purpose and scope: corrected conclusion

R-101 reduced the regular rational endpoint to

$$\mathcal{K}_R = G^T L_\Sigma c + \frac{1}{2} c^T \overline{B}_\Sigma (U + a) c, \quad G = DU, \quad c = Da, \quad (1.1)$$

where $\overline{B}_\Sigma = P_\Sigma B_R$ and L_Σ is the complete second-order Taylor remainder. The proposed next move was to recombine the base cubic and balanced remainder, use the bounded second Hessian, and attach R-096 predictable-baseline support collapse to the global R-086 current.

The recombination and bounded-Hessian observations are correct. The direct R-096 attachment is not: passage from a predictable insertion to the global R-101 current leaves a genuine future-feedback innovation. The decisive repair is to keep that innovation, interchange the finite $j < k$ sums, and condition the complete product at $\mathcal{F}_{k-1}$. Future heat then turns each column into one predictable physical-shell insertion. Pathwise shell estimates and one post-sum Young allocation give a cutoff- and heat-uniform absolute bound for the whole residual with slack $1/14$. This closes the regular one-shot K_R lower form while leaving its entire terminal square unused. Complete H_N , REG, and progression are still not claimed.

This note records seven rigorous advances.

1. It corrects the owner ledger: the complete L_Σ may be estimated as one Hessian secant only after refunding the provisional separate R-076 base-cubic charge. The R-076 theorem remains valid in its own owner.
2. It proves a target-heat-uniform bound on $D^2 P_\Sigma B_R$ for every positive-semidefinite covariance, including singular covariances.
3. It gives an exact one-dimensional Laplace–Wick representation of the whole heat-lifted production rational Gram for every $\Sigma \succeq 0$, and derives its isotropic scalar ridge expansion.
4. It proves active-subspace coercivity, full-rank pointwise positivity, and a kernel asymptotic which rules out global uniform inverse-Gram coercivity.
5. It derives the exact global-current cross-Doob decomposition, isolates the future-feedback residual, and falsifies the naive local-to-global bridge by an exact finite-mode fixture.
6. It proves the sharp deterministic-metric global Doob–Bessel reduction and records why its norm does not match the random terminal Gram reserve.
7. It closes the complete future-feedback residual on the regular mutually orthogonal strict-past one-shot class by swapping the root sums before estimation. The bound uses one X/Y allocation, is uniform in cutoff and all deterministic PSD target heats, and preserves the entire terminal square.

The scope is finite cutoff and fixed positive density floor. The exact heat formulas allow all finite-dimensional $\Sigma \succeq 0$. The stochastic owner and closure statements retain the R-101 locality, real-even value–gradient independence, deterministic target heat, and regular mutually orthogonal strict-past no-revisit physical-shell class. No arbitrary progressive/revisit control, cutoff or floor removal, Nelson estimate, interacting measure, or Sector-A closure is proved. Tier stays T4.

The binding authorities are R-050 and R-071 for canonical smooth one-form currents; R-075 and R-092 (6.8)–(6.9) for pathwise annular-shell and prefix budgets; R-076 and R-078 for the historical input-maximum and pure- U coordinates; R-080 for the fixed-low owner; R-086 for the exact earlier payable/coefficient-dominant split; R-096 for predictable-baseline support; R-099 for causal last-root ownership; and R-100 plus R-101 for the complete raw-Wick, heat, current, and square owners. In the active closure coordinate, the earlier separate R-086 $T_G^{\leq}$ current allocation is refunded rather than charged again; the R-086 theorem itself remains valid.

2 Full-Hessian owner recombination

For a smooth symmetric matrix field put

$$\begin{aligned} L_\Sigma(u, a) &= \bar{B}_\Sigma(u + a) - \bar{B}_\Sigma(u) - D\bar{B}_\Sigma(u)[a] - \frac{1}{2}D^2\bar{B}_\Sigma(u)[a, a] \\ &= \int_0^1 (1-t)\{D^2\bar{B}_\Sigma(u + ta) - D^2\bar{B}_\Sigma(u)\}[a, a] dt. \end{aligned} \quad (2.1)$$

It also has the exact split

$$L_\Sigma = \frac{1}{6}D^3\bar{B}_\Sigma(u)[a, a, a] + L_{\text{bal}, \Sigma}. \quad (2.2)$$

Theorem 2.1 (nonduplicating full-Hessian owner)

Inside the R-101 endpoint (1.1), using the complete secant (2.1) as one owner is legal only after refunding the separate R-076 base-cubic allocation. If that allocation is retained, only its complementary remainder may be charged. The complete-secant choice does not restore $L : Q/2$. That raw-Wick Taylor member was already terminalised by R-101 and remains unavailable.

Proof. Equation (2.2) is Taylor’s theorem, so its two summands add to the same $G^T L_\Sigma c$ owner in (1.1). Charging the first summand separately and then charging (2.1) counts it twice. Conversely, refunding the separate charge changes no algebra and invalidates no R-076 theorem. R-101 Theorem 6.1 assigns $L : Q/2$ to the complete raw-Wick block before the current split; changing the current coordinate does not reverse that assignment. $\square$

This correction matters numerically. On the normalized scalar ridge used below, the complete, cubic, and balanced coefficients are respectively

$$\frac{40}{81}, \quad -\frac{25}{81}, \quad \frac{65}{81}. \quad (2.3)$$

The full coefficient is not an additional reserve on top of the last two.

3 Uniform second-Hessian theorem

Let $\Pi = P_{\text{dbl}}$, $e = \epsilon_\rho = 10^{-12}$, $p_0 = 4 + e$, $c_1 = 243/(8000p_0)$, and $\alpha = 5/9$. The production rational row and Gram are

$$g(z) = \Pi z - \alpha \frac{|\Pi z|^2}{e + |z|^2} z, \quad B_R(z) = 4c_1 g(z) \otimes g(z). \quad (3.1)$$

Set

$$\rho = (e + |z|^2)^{1/2}, \quad n = \frac{z}{\rho}, \quad h(n) = \Pi n - \alpha(n^T \Pi n)n, \quad H = h \otimes h. \quad (3.2)$$

Then

$$g(z) = \rho h(n), \quad B_R(z) = 4c_1 \rho^2 H(n). \quad (3.3)$$

Lemma 3.1 (finite normalized Hessian)

For $R_n = I - nn^T$ and a direction v , define

$$A_v(n) = 2(n \cdot v)H(n) + DH(n)[R_n v]. \quad (3.4)$$

Then, for every pair of directions v, w ,

$$\boxed{D^2\{\rho^2 H(n)\}[v, w] = (n \cdot w)A_v(n) + DA_v(n)[R_n w].} \quad (3.5)$$

Consequently

$$M_2 := \sup_z \|D^2 B_R(z)\| < \infty. \quad (3.6)$$

Proof. The elementary derivatives are

$$D\rho[v] = n \cdot v, \quad Dn[v] = \frac{R_n v}{\rho}. \quad (3.7)$$

They give $D(\rho^2 H)[v] = \rho A_v(n)$; one more derivative gives (3.5). The map h is a polynomial in n , so the right side of (3.5) is a finite polynomial on the closed unit ball $|n| \leq 1$. It is therefore bounded. $\square$

Corollary 3.2 (all-PSD heat contraction)

For every positive-semidefinite target covariance,

$$D^2 P_\Sigma B_R = P_\Sigma D^2 B_R, \quad \sup_z \|D^2 P_\Sigma B_R(z)\| \leq M_2. \quad (3.8)$$

In particular,

$$\boxed{|L_\Sigma(u, a)| \leq M_2 |a|^2} \quad (3.9)$$

uniformly in $\Sigma \succeq 0$.

Proof. The bounded second derivative permits differentiation under Gaussian convolution, and averaging is an L^∞ contraction. The integrand in (2.1) is bounded by $2M_2|a|^2$; integration of $1 - t$ gives (3.9). $\square$

This is a pointwise tensor theorem, not a multiplier theorem. In the earlier R-086 split coordinate, (3.9) preserves the historical payload-comparable estimate

$$|T_G^\leq| \lesssim M_G X^{2/5} Y^{11/30}, \quad (3.10)$$

and the one-use Young bound

$$M_G X^{2/5} Y^{11/30} \leq \eta X + \zeta Y + C \eta^{-12/7} \zeta^{-11/7} M_G^{30/7}. \quad (3.11)$$

The exponent slack is $7/30$. It does not control $T_G^>$ in that coordinate, where high adapted coefficient frequencies resonate with the high Gaussian current. The active Theorem 9.2 estimates the complete current after chronological reindexing, so this separate $T_G^\leq$ charge is refunded.

The spatial audit sharpens that boundary. R-078 already reconstructs the pure high- U principal through $U \circ DU$; at $\varepsilon = 1/10$ its powers are

$$X^{11/40}Y^{49/120}, \quad \text{slack} = \frac{19}{60}, \quad p = \frac{60}{19}. \quad (3.12)$$

Under the split alternative, the coefficient-low/high- a principal is the available R-076 a^3Da input-maximum payload,

$$X^{2/5}Y^{8/15}, \quad \text{slack} = \frac{1}{15}, \quad p = 15. \quad (3.13)$$

It is historical support for that coordinate and is not charged in the active complete- L ledger used below. The surviving high- a /mixed commutator has the absolute constraint

$$q > 1 + s, \quad 1 - \frac{q}{2} - \frac{4-q}{6} = \frac{1-q}{3} < -\frac{s}{3}. \quad (3.14)$$

At $s = 3/5$ it has negative slack below $-1/5$. Equivalently, isolating the balanced hidden control produces $X^{2/5}Y^{7/10}$, total $11/10$; a separated multiplier route has total $13/12$. These failures identify the innovation branch below; they are not counterexamples to the signed paid form.

4 All-covariance Laplace–Wick representation

Let $h_\Sigma \sim N(0, \Sigma)$, $Y = z + h_\Sigma$, and retain

$$p_0 = 4 + e, \quad c_1 = \frac{243}{8000p_0}. \quad (4.1)$$

For $t \geq 0$ define

$$A_t = I + 2t\Sigma, \quad m_t = A_t^{-1}z, \quad C_t = \Sigma A_t^{-1}, \quad (4.2)$$

$$\Psi_t(z, \xi) = \det(A_t)^{-1/2} \exp \left\{ -tz^T A_t^{-1}z + \xi^T m_t + \frac{1}{2} \xi^T C_t \xi \right\}. \quad (4.3)$$

For $k = 1, 2$, put

$$\mathfrak{W}_k(t, z) = (\nabla_\xi \otimes \nabla_\xi)(\nabla_\xi^T \Pi \nabla_\xi)^k \Psi_t(z, \xi) \Big|_{\xi=0}. \quad (4.4)$$

Theorem 4.1 (exact all-PSD rational heat formula)

For every finite-dimensional $\Sigma \succeq 0$,

$$\mathfrak{W}_k(t, z) = \mathbb{E} \left[e^{-t|Y|^2} (Y^T \Pi Y)^k Y \otimes Y \right], \quad (4.5)$$

and

$$\boxed{\begin{aligned} (P_\Sigma B_R)(z) &= \frac{243}{2000p_0} \Pi(zz^T + \Sigma) \Pi \\ &\quad - \frac{27}{400p_0} \int_0^\infty e^{-et} \{ \Pi \mathfrak{W}_1 + \mathfrak{W}_1 \Pi \} dt \\ &\quad + \frac{3}{80p_0} \int_0^\infty t e^{-et} \mathfrak{W}_2 dt. \end{aligned}} \quad (4.6)$$

Proof. Completing the Gaussian square gives (4.3) for $\mathbb{E}e^{-t|Y|^2+\xi^TY}$, without inverting Σ . Differentiating in ξ proves (4.5), including singular Σ . With $r = Y^T\Pi Y$ and $d = e + |Y|^2$,

$$\begin{aligned} g(Y) \otimes g(Y) &= (\Pi Y) \otimes (\Pi Y) - \alpha \frac{r}{d} \{\Pi Y \otimes Y + Y \otimes \Pi Y\} \\ &\quad + \alpha^2 \frac{r^2}{d^2} Y \otimes Y. \end{aligned} \quad (4.7)$$

Use

$$d^{-1} = \int_0^\infty e^{-td} dt, \quad d^{-2} = \int_0^\infty te^{-td} dt, \quad (4.8)$$

then multiply by $4c_1$. The three derived coefficients are $243/(2000p_0)$, $-27/(400p_0)$, and $3/(80p_0)$. $\square$

At fixed $e > 0$, every required z -derivative along the compact Taylor segment is dominated by a Gaussian polynomial moment times $C(1+t)^N e^{-et}$. This proves absolute convergence and justifies the differentiation and Taylor-remainder interchange used below.

For the Taylor operator

$$\mathcal{R}_q^a F(z) = F(z+a) - \sum_{\ell=0}^q \frac{1}{\ell!} D^\ell F(z)[a^\ell], \quad (4.9)$$

the polynomial first line of (4.6) vanishes when $q \geq 2$. Hence

$$\begin{aligned} L_\Sigma = \mathcal{R}_2^a P_\Sigma B_R &= -\frac{27}{400p_0} \int_0^\infty e^{-et} \mathcal{R}_2^a \{\Pi \mathfrak{W}_1 + \mathfrak{W}_1 \Pi\} dt \\ &\quad + \frac{3}{80p_0} \int_0^\infty te^{-et} \mathcal{R}_2^a \mathfrak{W}_2 dt. \end{aligned} \quad (4.10)$$

The same formula with $\mathcal{R}_3^a$ gives the balanced remainder. Formula (4.10) removes the rational denominator analytically, but it creates no physical-space smoothing and supplies no sign for $G^T L_\Sigma c$.

5 Isotropic scalar ridge

Take dimension six, rank $\Pi = 4$, $\Sigma = \sigma I_6$, and $z = xe_1$ with $\Pi e_1 = e_1$. Put

$$s_t = 1 + 2\sigma t, \quad \lambda_t = \frac{t}{s_t}. \quad (5.1)$$

Gaussian moment contraction gives

$$F_1(t, x) = e^{-\lambda_t x^2} \left(\frac{x^4}{s_t^7} + \frac{9\sigma x^2}{s_t^6} + \frac{6\sigma^2}{s_t^5} \right), \quad (5.2)$$

$$F_2(t, x) = e^{-\lambda_t x^2} \left(\frac{x^6}{s_t^9} + \frac{21\sigma x^4}{s_t^8} + \frac{96\sigma^2 x^2}{s_t^7} + \frac{48\sigma^3}{s_t^6} \right). \quad (5.3)$$

Thus, for $\bar{b}_\sigma(x) = e_1^T (P_{\sigma I_6} B_R)(xe_1)e_1$,

$$\boxed{\bar{b}_\sigma(x) = 4c_1 \left[x^2 + \sigma - \frac{10}{9} \int_0^\infty e^{-et} F_1 dt + \frac{25}{81} \int_0^\infty te^{-et} F_2 dt \right]}. \quad (5.4)$$

At $u = \sqrt{e}$, $a = -\sqrt{e}$, define

$$L_\sigma = \bar{b}_\sigma(0) - \bar{b}_\sigma(\sqrt{e}) + \sqrt{e} \bar{b}'_\sigma(\sqrt{e}) - \frac{e}{2} \bar{b}''_\sigma(\sqrt{e}),$$

$$L_{\text{cubic},\sigma} = -\frac{e^{3/2}}{6} \bar{b}_\sigma'''(\sqrt{e}), \quad L_{\text{bal},\sigma} = L_\sigma - L_{\text{cubic},\sigma}.$$

Then, as $\sigma/e \downarrow 0$,

$$\bar{b}_\sigma(0) = 4c_1\sigma - \frac{80}{3}c_1\frac{\sigma^2}{e} + O\left(c_1\frac{\sigma^3}{e^2}\right), \quad (5.5)$$

$$L_\sigma = \frac{40}{81}c_1e + \frac{65}{9}c_1\sigma + O\left(c_1\frac{\sigma^2}{e}\right), \quad (5.6)$$

$$L_{\text{cubic},\sigma} = -\frac{25}{81}c_1e + \frac{25}{3}c_1\sigma + O\left(c_1\frac{\sigma^2}{e}\right), \quad (5.7)$$

$$L_{\text{bal},\sigma} = \frac{65}{81}c_1e - \frac{10}{9}c_1\sigma + O\left(c_1\frac{\sigma^2}{e}\right). \quad (5.8)$$

Writing $s = x/\sqrt{e}$, an independent derivation uses the exact first heat generator

$$\frac{1}{2}\Delta_6 B_{11}(\sqrt{e}se_1) = \frac{4c_1(32s^8 + 68s^6 - 53s^4 - 162s^2 + 27)}{27(1+s^2)^4}. \quad (5.9)$$

Equations (5.6)–(5.8) confirm the owner refund in Theorem 2.1. They do not promote one scalar ray to an all-mode sign theorem.

The exact scalar square completion still gives

$$\inf_c \left\{ GL_{\text{bal},\sigma}c + \frac{\theta}{2}\bar{b}_\sigma(0)c^2 \right\} = -\frac{G^2 L_{\text{bal},\sigma}^2}{2\theta\bar{b}_\sigma(0)} \quad (5.10)$$

and therefore the leading loss

$$-\frac{4225}{52488\theta} \frac{c_1 e^2}{\sigma} G^2. \quad (5.11)$$

This rules out an endpoint-square-only estimate uniform down to zero heat. It is not a counterexample once Cameron–Martin energy and the other owners are retained.

6 Active and kernel geometry

Theorem 6.1 (active coercivity and kernel nonuniformity)

For $q \in \text{Ran } \Pi$,

$$\boxed{\frac{64c_1}{81} \{(q \cdot z)^2 + q^T \Sigma q\} \leq q^T (P_\Sigma B_R)(z) q \leq 4c_1 \{(q \cdot z)^2 + q^T \Sigma q\}.} \quad (6.1)$$

If Σ is positive definite, then $P_\Sigma B_R(z)$ is positive definite at every z . Nevertheless, for $\sigma > 0$, a unit $k \in \ker \Pi$, and $\Sigma = \sigma I_6$,

$$\boxed{T^2 k^T (P_{\sigma I_6} B_R)(Tk) k \xrightarrow{T \rightarrow \infty} \frac{800}{27} c_1 \sigma^2.} \quad (6.2)$$

Proof. For active q ,

$$q \cdot g(Y) = \left(1 - \alpha \frac{|\Pi Y|^2}{e + |Y|^2}\right) q \cdot Y, \quad (6.3)$$

and the parenthesis lies in $[4/9, 1]$. Squaring and averaging proves (6.1). If $c^T P_\Sigma B_R(z) c = 0$ for full-rank heat, analyticity and full Gaussian support force $c \cdot g(y) = 0$ for every y . Testing active y first gives $\Pi c = 0$; a mixed active/kernel y then gives $c = 0$.

For (6.2), write $Y = Tk + h$. Then $Tk \cdot g(Y) \rightarrow -\alpha|\Pi h|^2$. Since $\mathbb{E}|\Pi h|^4 = 4(4+2)\sigma^2 = 24\sigma^2$, multiplication by $4c_1$ gives $800c_1\sigma^2/27$. $\square$

Thus fixed full-rank heat fills the pointwise kernel but does not give a field-uniform smallest eigenvalue. Active coercivity cannot be applied to a general c by discarding its kernel component or the Gram cross block.

7 Exact global-current cross-Doob decomposition

Let $P_j = \mathbb{E}[\cdot | \mathcal{F}_j]$, $d_j = P_j - P_{j-1}$, and $G_j = P_j G_J$. Define the terminal current coefficient

$$\Psi_* = L_{\Sigma_{\text{tar}}}(U_J, A^*)c^*, \quad c^* = DA^*, \quad F_j = P_j \Psi_*. \quad (7.1)$$

Theorem 7.1 (future-feedback residual identity)

For every fixed low level ℓ ,

$$\mathbb{E}\langle G_J, \Psi_* \rangle = \mathbb{E}\langle G_\ell, F_\ell \rangle + \sum_{j>\ell} \mathbb{E}\langle d_j G_J, d_j F_j \rangle. \quad (7.2)$$

For partial controls, put

$$\Psi^{(j)} = L_{\Sigma_{\text{tar}}}(U_J, A^{(j)})c^{(j)}, \quad \Lambda_j = P_j \Psi^{(j)}. \quad (7.3)$$

Under the R-101 locality, real-even value–gradient independence, and strict-past hypotheses,

$$\mathbb{E}\langle d_j G_J, \Lambda_j \rangle = 0. \quad (7.4)$$

Consequently the exact survivor is

$$\mathcal{R}_{\text{cur}} = \sum_{j>\ell} \mathbb{E} \left\langle d_j G_J, P_j (\Psi_* - \Psi^{(j)}) \right\rangle. \quad (7.5)$$

If $\Delta_k \Psi^{(k)} = \Psi^{(k)} - \Psi^{(k-1)}$, then, with that sign convention,

$$\mathcal{R}_{\text{cur}} = \sum_{\ell < j < k \leq J} \mathbb{E} \langle d_j G_J, \Delta_k \Psi^{(k)} \rangle. \quad (7.6)$$

Proof. Both G_j and F_j are Hilbert-valued martingales, so their cross-Doob increments prove (7.2). The local partial coefficient in (7.3) depends on the fresh root only through its value coordinate; $c^{(j)}$ is predictable. The fresh derivative coordinate is independent at the same point, proving (7.4). Since $d_j G_J$ centres every $\mathcal{F}_{j-1}$ quantity, subtracting Λ_j from the j th bracket gives (7.5). Telescope $\Psi_* - \Psi^{(j)}$ over $k > j$ and use conditional Fubini to obtain (7.6). $\square$

The product structure cannot be suppressed. For a bilinear multiplication $m(H, c) = Hc$, the exact difference is

$$\begin{aligned} d_j m(H, c) &= m(d_j H, P_{j-1} c) + m(P_{j-1} H, d_j c) + m(d_j H, d_j c) \\ &\quad + \text{Cov}_j^m(H, c) - \text{Cov}_{j-1}^m(H, c). \end{aligned} \quad (7.7)$$

Here $\text{Cov}_j^m(H, c) = P_j m(H, c) - m(P_j H, P_j c)$. R-101 centres a coefficient against a fresh raw-Wick derivative increment; it does not delete $d_j c$, the product increment, or the covariance defect in (7.7). R-096 applies to each predictable baseline after complete cancellation, but (7.5)–(7.7) are precisely what is lost when that local support result is identified with the global R-101 current.

8 Finite-mode bridge and Cartan counterfixtures

On the active scalar ray with $e = 1$ and c_1 removed, the Gram is

$$b(x) = \frac{4x^2(4x^2 + 9)^2}{81(1 + x^2)^2}. \quad (8.1)$$

Define

$$\ell(x, A) = b(x + A) - b(x) - b'(x)A - \frac{1}{2}b''(x)A^2, \quad (8.2)$$

$$\ell_{\text{bal}}(x, A) = \ell(x, A) - \frac{1}{6}b'''(x)A^3. \quad (8.3)$$

Exact differentiation gives

$$b'(1) = \frac{208}{81}, \quad b''(1) = -\frac{2}{81}, \quad b'''(1) = \frac{50}{27}, \quad (8.4)$$

$$\partial_A \ell(1, -1) = -\frac{70}{27}, \quad \partial_A \ell_{\text{bal}}(1, -1) = -\frac{95}{27}. \quad (8.5)$$

Proposition 8.1 (naive bridge is false)

Let $\xi \sim N(0, 1)$ be an early derivative innovation and let a later strict-past control produce $A = -1 + \lambda\xi$. Then

$$\boxed{\mathbb{E}[\xi \ell(1, A)] = -\frac{70}{27}\lambda} \quad (8.6)$$

exactly, while

$$\boxed{\mathbb{E}[\xi \ell_{\text{bal}}(1, A)] = -\frac{95}{27}\lambda - \frac{25}{27}\lambda^3}. \quad (8.7)$$

Proof. The endpoint term $b(\lambda\xi)$ is even, so its product with ξ has zero expectation. The remaining terms in (8.2) use only the first two Gaussian moments and give (8.6). Subtracting the cubic in (8.3), with $\mathbb{E}\xi^4 = 3$, gives (8.7). $\square$

For an exact two-root finite filtration, take independent standard Gaussian roots ξ_1, ξ_2 , terminal coefficient $\psi = \ell(1, -1 + \lambda\xi_1)$, and $G_0 = 2$, $G_1 = 2 + \xi_1$, $G_2 = 2 + \xi_1 + \xi_2$. With $F_j = \mathbb{E}_j\psi$, the fixed-low term $\mathbb{E}(G_0 F_0)$ is nonzero, the first innovation bracket is $-70\lambda/27$, and the second is zero. Thus the full cross-Doob identity retains both the low endpoint and a nonzero future-feedback bracket; the predictable constant baseline pairs to zero.

Proposition 8.2 (future-control increments do not determine the nonlinear current)

Let $\xi \sim N(0, 1)$ and let δ be an independent Rademacher sign. Use the filtration $\mathcal{F}_{j-1}$ trivial, $\mathcal{F}_j = \sigma(\xi)$, $\mathcal{F}_{j+1} = \sigma(\xi, \delta)$, and insert at $k = j + 2$ the strict-past scalar control and tangent

$$t = \frac{1}{10}, \quad q = \frac{1}{2}, \quad a = c_a = t\delta(1 + q \operatorname{sgn} \xi). \quad (8.8)$$

Put $G_{j-1} = 0$, $G_j = G_J = -\xi$, and $\Psi(a) = \ell(1, a)a$. Conditional averaging over δ gives

$$d_j a = d_j c_a = 0. \quad (8.9)$$

Nevertheless, with $H(r) = \{\Psi(r) + \Psi(-r)\}/2$,

$$H(r) = \frac{-40r^4(2r^6 + 13r^2 - 10)}{81(r^2 - 2r + 2)^2(r^2 + 2r + 2)^2}, \quad (8.10)$$

and exact rational arithmetic gives

$$H(3/20) - H(1/20) = \frac{10175618597178586187512}{67965137546788211215457205} > 0. \quad (8.11)$$

Since $P_j \Psi = H(t(1 + q \operatorname{sgn} \xi))$,

$$\boxed{\mathcal{R}_{\text{cur}} = -\sqrt{2/\pi}, \frac{5087809298589293093756}{67965137546788211215457205} < 0.} \quad (8.12)$$

Thus even vanishing martingale differences of the future control and its tangent do not determine the nonlinear coefficient innovation. This is a method no-go only: the terminal square, X/Y payments, heat, trace, forest, and production companions are absent, so (8.12) does not refute (9.1).

For comparison, an abstract real-even bilinear frequency diagnostic uses

$$c_r = h \cos(rx), \quad d_m = \gamma \{ \xi \cos(mx) - \eta \sin(mx) \}, \quad (8.13)$$

and a later shift

$$a_{m+r} = \sqrt{e} \lambda \xi \cos((m+r)x). \quad (8.14)$$

Since

$$\langle \cos(mx) \cos((m+r)x) \cos(rx) \rangle = \frac{1}{4}, \quad (8.15)$$

the selected full bilinear coefficient is

$$-\frac{35}{54} c_1 e \gamma h \lambda, \quad (8.16)$$

and the selected balanced bilinear coefficient is

$$-\frac{95}{108} c_1 e \gamma h \lambda + O(\lambda^3). \quad (8.17)$$

Equations (8.13)–(8.17) verify only a scalar-fibre coefficient and carrier overlap. They do not impose a complete production control $c = Da$, restore the deterministic ridge background, or include the same-order value/derivative companion. They are therefore a frequency diagnostic, not a production-current fixture and not evidence for deleting or retaining (7.5). The exact two-root calculation above is the bridge no-go.

There is also a field-space Cartan obstruction. On the two-dimensional active/inactive slice, put

$$g(x, y) = \left(x - \alpha \frac{x^3}{d}, -\alpha \frac{x^2 y}{d} \right), \quad d = 1 + x^2 + y^2, \quad B = 4g \otimes g. \quad (8.18)$$

For $a = c = e_1$, let

$$L = B(z + a) - B(z) - DB(z)[a] - \frac{1}{2} D^2 B(z)[a, a], \quad \omega = Lc. \quad (8.19)$$

At $z = (1, 1)$,

$$\boxed{\partial_y \omega_x - \partial_x \omega_y = -\frac{40}{729} \neq 0.} \quad (8.20)$$

Thus even the complete full- L current one-form is not a field-space gradient. Chain-rule integration can split off an exact part but cannot delete the genuine Cartan/enhanced-current remainder.

Neither (8.16), (8.12), nor (8.20) is a paid-form counterexample. The endpoint square, Cameron–Martin energy, trace, forest, low term, and other production rows are intentionally absent.

9 Exact successor and proof map

Theorem 9.1 (strongest deterministic-metric Doob reduction)

Let $Q_\ell = I - P_\ell$. The exact identity (7.2) can equivalently be written

$$\boxed{\mathcal{R}_{\text{cur}} = \mathbb{E}\langle Q_\ell G_J, Q_\ell \Psi_* \rangle.} \quad (9.0)$$

Consequently, for every deterministic positive spatial multiplier W for which the displayed terms are defined and every $\theta > 0$,

$$\boxed{\mathcal{R}_{\text{cur}} + \frac{\theta}{2} \mathbb{E} \|W Q_\ell \Psi_*\|^2 \geq -\frac{1}{2\theta} \mathbb{E} \|W^{-1} Q_\ell G_J\|^2.} \quad (9.0a)$$

Indeed, martingale orthogonality proves (9.0), and (9.0a) is the Hilbert-space completion of the square after all roots have been summed. It is therefore the sharp generic Doob/Bessel reduction and avoids rootwise reuse of an energy budget.

This theorem alone does not prove the production lower form below. Its positive norm is the innovation of $\Psi_* = L_{\Sigma_{\text{tar}}} c^*$ in a deterministic metric, whereas the available endpoint reserve is the random, possibly singular $B_1 = (P_{\Sigma_{\text{tar}}} B_R)(U_J + A^*)$ square. No accepted result controls the inverse-metric norm on the right of (9.0a) by the one-use Cameron–Martin budget. R-099 closes factorized linear future-tail subchannels, but the complete rational response does not factor through one predictable derivative increment; R-081’s Jensen blind spot already permits vanishing partial coefficient derivatives with a surviving nonlinear coefficient innovation. Thus a production-specific random-metric, product-covariance, square–trace–forest estimate would be required if one stayed in this coordinate. The next theorem instead returns the complete product to its last future insertion before estimating it.

For reference, the sufficient regular rational form is: for every fixed $0 < \theta < 1$,

$$\boxed{\mathcal{R}_{\text{cur}} + \frac{\theta}{2} \mathbb{E} \sum_i \int (D_i A^*)^T (P_{\Sigma_{\text{tar}}} B_R)(U_J + A^*) D_i A^* dx \geq -\eta \mathbb{E} X - \zeta \mathbb{E} Y - C_{\eta, \zeta, \theta, e, j_0}.} \quad (9.1)$$

The proof must satisfy all of the following.

1. The constant is uniform in terminal cutoff and in the declared production family of positive-semidefinite target covariances, including its singular boundary if that boundary occurs in the production closure.
2. The sum over $j < k$, spatial blocks, and product-covariance companions is performed before one global Young inequality. No rootwise reuse of X or Y is allowed.
3. The fixed-low endpoint, heat, trace, R-063 forest, and R-100 residual remain in their existing owners exactly once.
4. The unused terminal-square fraction is reserved for the later full-frame assembly.
5. No inverse-Gram bound, predictable BMO estimate, or Malliavin derivative of an admissible control is assumed.

Theorem 9.2 (regular future-insertion current closure)

Assume the regular mutually orthogonal strict-past no-revisit physical-shell class. Write

$$A^{(k)} = A^{(\ell)} + \sum_{r=\ell+1}^k a_r, \quad c^{(k)} = D A^{(k)}, \quad a_k = K_k h_k, \quad h_k \in \mathcal{F}_{k-1}, \quad (9.2)$$

and assume that the target and future heat covariances are deterministic PSD increments. Then, for every $\eta, \zeta > 0$,

$$|\mathcal{R}_{\text{cur}}| \leq \eta \mathbb{E}X + \zeta \mathbb{E}Y + C_{\eta, \zeta, e, j_0}, \quad (9.3)$$

uniformly in the terminal cutoff and in the deterministic PSD target heat. Consequently (9.1) holds with the entire terminal square unspent.

Exact last-insertion identity. Finite Fubini in (7.6) gives

$$\mathcal{R}_{\text{cur}} = \sum_{k > \ell} \mathbb{E} \langle G_{k-1} - G_\ell, \Delta_k \Psi^{(k)} \rangle. \quad (9.4)$$

Put $\widehat{G}_{k-1} = G_{k-1} - G_\ell$ and

$$\widetilde{L}_k = L_{\Sigma_{\text{tar}} + \Sigma_{[k; J]}}.$$

Both $A^{(k)}$ and $c^{(k)}$ are $\mathcal{F}_{k-1}$ -measurable. Conditioning the *whole product* in (9.4), then using heat composition and commutation of the value derivative with deterministic heat, yields the exact split

$$\boxed{\begin{aligned} \mathcal{R}_{\text{cur}} &= \mathcal{R}_D + \mathcal{R}_C, & \mathcal{R}_D &= \sum_{k > \ell} \mathbb{E} \langle \widehat{G}_{k-1}, \widetilde{L}_k(U_{k-1}, A^{(k)}) D a_k \rangle, \\ \mathcal{R}_C &= \sum_{k > \ell} \mathbb{E} \langle \widehat{G}_{k-1}, \{ \widetilde{L}_k(U_{k-1}, A^{(k)}) - \widetilde{L}_k(U_{k-1}, A^{(k-1)}) \} c^{(k-1)} \rangle. \end{aligned}} \quad (9.5)$$

No product covariance or Jensen term is discarded: both are already inside the conditioned complete-product difference in (9.5).

From (2.1) and Corollary 3.2,

$$|\widetilde{L}_k(u, A)| \leq M_2 |A|^2, \quad |\widetilde{L}_k(u, A + a) - \widetilde{L}_k(u, A)| \leq 2M_2 \{|A||a| + |a|^2/2\}. \quad (9.6)$$

The second inequality follows by differentiating the integral remainder:

$$D_A \widetilde{L}_k(u, A)[h] = \int_0^1 \{D^2 \overline{B}(u + tA) - D^2 \overline{B}(u)\}[A, h] dt. \quad (9.7)$$

One-use shell ledger. Set

$$x_k = \|h_k\|_{\mathcal{H}_k}^2, \quad v_k = \|a_k\|_6, \quad R_k = 2^{-k/2} \|\widehat{G}_{k-1}\|_6. \quad (9.8)$$

Set, pathwise,

$$X = \sum_{k > \ell} x_k, \quad Y = \|U + A^*\|_6^6, \quad \overline{Y} = 1 + Y. \quad (9.8a)$$

Thus X is the accepted high-shell Cameron–Martin energy, not a terminal H^2 norm. The R-075/R-092 physical-shell bounds and the no-revisit identity give a cutoff-uniform Gaussian factor Γ_U with every fixed moment such that

$$\sum_k x_k = X, \quad \sum_k v_k^6 \lesssim \Gamma_U^6 \overline{Y}, \quad \|a_k\|_2 \lesssim 2^{-2k} x_k^{1/2}, \quad \|D a_k\|_2 \lesssim 2^{-k} x_k^{1/2}, \quad (9.9)$$

$$\sup_k \|A^{(k)}\|_6 \lesssim \Gamma_U \overline{Y}^{1/6}, \quad \sup_k \|D(A^{(k)} - A^{(\ell)})\|_p \lesssim X^{1/2}, \quad 2 \leq p \leq 6. \quad (9.10)$$

The full derivative is not silently replaced by the high prefix. Write

$$c^{(k)} = c_{\text{low}} + c_{\text{hi}}^{(k)}, \quad c_{\text{low}} = D A^{(\ell)}.$$

R-080's fixed-low Bernstein estimate and $A^{(\ell)} = P_{<j_0}(U + A^*) - P_{<j_0}U$ give another all-moment Gaussian factor Γ_0 such that

$$\|c_{\text{low}}\|_p \lesssim \Gamma_0 \bar{Y}^{1/6}, \quad \|c_{\text{hi}}^{(k)}\|_p \lesssim X^{1/2}. \quad (9.10b)$$

The two R-080 low boundary objects remain in their own owners; (9.10b) only keeps the low derivative that actually occurs in the present mixed current.

Take absolute values in (9.5), move the triangle inequality inside each expectation, and only then apply the following pathwise shell estimates. Spatial Hölder with $(6, 3, 2)$ and shell Cauchy give

$$|\mathcal{R}_D| \lesssim M_2 \mathbb{E}[\widehat{\mathcal{Z}}_D X^{1/2} \bar{Y}^{1/3}], \quad \mathcal{Z}_D = \left(\sum_k 2^{-k} R_k^2 \right)^{1/2}, \quad \widehat{\mathcal{Z}}_D = \Gamma_U^2 \mathcal{Z}_D. \quad (9.11)$$

For the $|A^{(k-1)}| |a_k| |c^{(k-1)}|$ part of $\mathcal{R}_C$, use spatial exponents $(6, 6, 42/11, 42/17)$ and

$$\|a_k\|_{42/11} \leq \|a_k\|_2^{2/7} \|a_k\|_6^{5/7}.$$

For the $|a_k|^2 |c^{(k-1)}|$ part, use $(6, 14/3, 14/3, 42/17)$ and

$$\|a_k\|_{14/3} \leq \|a_k\|_2^{1/7} \|a_k\|_6^{6/7}.$$

Both branches have the exact shell factor $2^{k/2} 2^{-4k/7} = 2^{-k/14}$. Hölder in k , followed by the split (9.10b), gives the complete bound

$$|\mathcal{R}_C| \lesssim M_2 \mathbb{E}[\widehat{\mathcal{Z}}_H X^{9/14} \bar{Y}^{2/7} + \widehat{\mathcal{Z}}_L X^{1/7} \bar{Y}^{19/42}]. \quad (9.12)$$

The second summand is the fixed-low derivative branch; omitting it would be incorrect. Here

$$\mathcal{Z}_1 = \left(\sum_k 2^{-3k/31} R_k^{42/31} \right)^{31/42}, \quad \mathcal{Z}_2 = \left(\sum_k 2^{-k/8} R_k^{7/4} \right)^{4/7}, \quad (9.13)$$

$$\widehat{\mathcal{Z}}_H = \Gamma_U^{12/7} (\mathcal{Z}_1 + \mathcal{Z}_2), \quad \widehat{\mathcal{Z}}_L = \Gamma_0 \widehat{\mathcal{Z}}_H.$$

The normalized Gaussian prefixes, Γ_U , and Γ_0 have every fixed moment. Minkowski, Cauchy–Schwarz, and the geometric weights give

$$\begin{aligned} \mathbb{E} \widehat{\mathcal{Z}}_D^6 &\lesssim 2^{-3j_0}, \\ \mathbb{E} \widehat{\mathcal{Z}}_H^{14} &\lesssim 2^{-j_0}, \\ \mathbb{E} \widehat{\mathcal{Z}}_L^{42/17} &\lesssim 2^{-3j_0/17}. \end{aligned} \quad (9.14)$$

These bounds are uniform in J . The three budget totals and Young slacks are

branch	budget total	slack
$\mathcal{R}_D$	$1/2 + 1/3 = 5/6$	$1/6$
$\mathcal{R}_{C,\text{hi}}$	$9/14 + 2/7 = 13/14$	$1/14$
$\mathcal{R}_{C,\text{low}}$	$1/7 + 19/42 = 25/42$	$17/42$

(9.15)

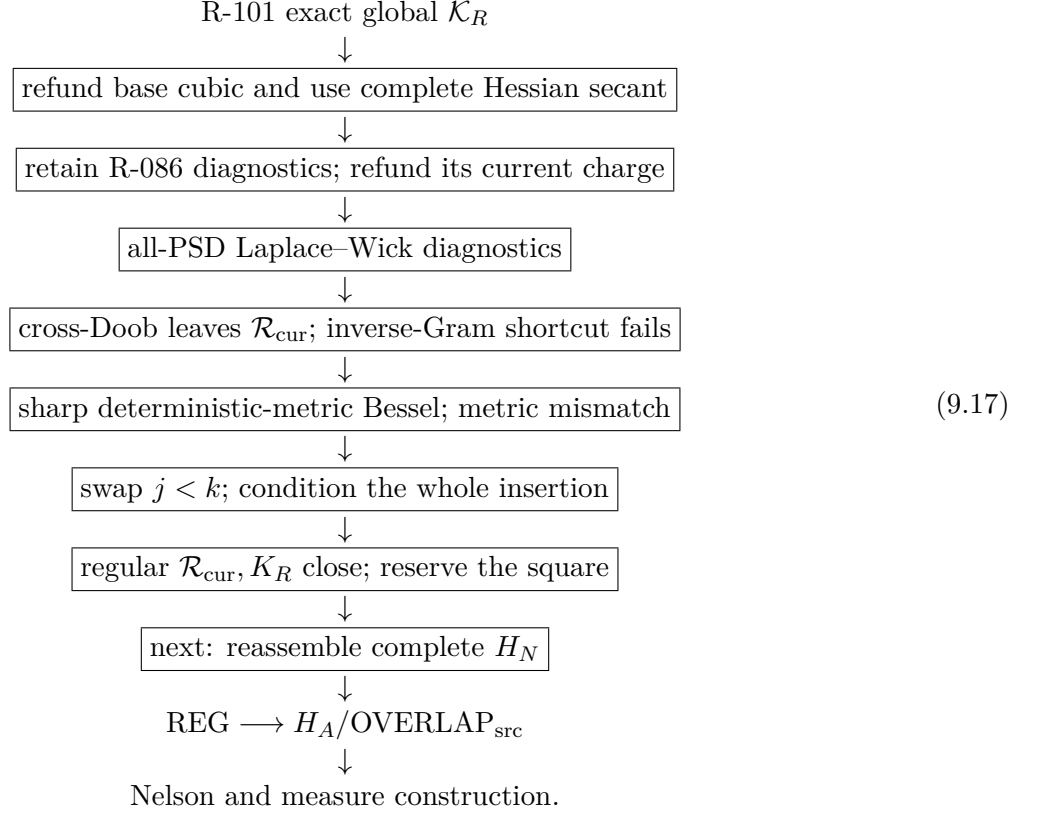
Applying generalized Young only after the complete sums in (9.11)–(9.12) proves (9.3). Besides the derivative remainder, the two coefficient remainders have the explicit forms

$$CM_2^{14} 2^{-j_0} \eta^{-9} \zeta^{-4} + CM_2^{42/17} 2^{-3j_0/17} \eta^{-6/17} \zeta^{-19/17}. \quad (9.16)$$

The first is dominant as the allocation parameters tend to zero. These are fixed Gaussian moments, not higher control moments. $\square$

Corollary 9.3 (regular K_R lower form). Combine (9.3) with the fixed-low owner already paid by R-080 and the nonnegative heat-lifted terminal square in (1.1). The regular one-shot K_R lower form is proved with one X/Y allocation and with the whole square reserved. The active coordinate therefore refunds the earlier separate R-086 $T_G^{\leq}$ current charge. This does not withdraw R-086, and it does not prove complete H_N , REG, or a progressive/revisit estimate.

The corrected proof map is



The route decisions are recorded in EXP-000250–EXP-000257. The final entry records and repairs the fixed-low derivative omission found during freeze review. The false global-to-predictable identification and the failed chain primitive are registered respectively as NG-2026-07-28-A13-GLOBAL-TO-PREDICTABLE-CURRENT-BRIDGE and NG-2026-07-28-A13-FULL-HESSIAN-CARTAN-CHAIN-PRIMITIVE.

10 Failed routes and reusable conclusions

Route	Verdict	Reusable conclusion
Complete Hessian recombination	advanced	Legal only after refunding the separate base-cubic charge; gives payload a^2Da and moment 30/7.
Bounded D^2B_R as multiplier	failed	It is a pointwise tensor bound; high adapted coefficient and Gaussian-current blocks may resonate to low frequency.
Pure high- U principal	closed	R-078 and $U \circ DU$ give slack 19/60 and moment 60/19.
Coefficient-low/high- a principal	split-only	R-076 input maximum gives slack 1/15 and moment 15 if the base cubic is retained; it is refunded in the active complete- L coordinate.
Absolute shifted commutator	failed	The exact slack is negative as in (3.14); the split balanced total is 11/10.
R-096 support implies global payment	failed	The cross-Doob passage leaves the future-feedback residual (7.5).
Deterministic-metric Doob/Bessel	boundary	Summing all roots before square completion gives the exact sharp bound (9.0a), but its deterministic norm and inverse-current cost do not match the random, possibly singular B_1 reserve.
Laplace–Wick positivity	boundary	Raw $\mathfrak{W}_2$ is PSD; its Taylor remainders and current are signless.
Future-control increment reduction	failed	The exact fixture (8.8)–(8.12) has $d_ja = d_jc_a = 0$ but a negative nonlinear current.
Full- L chain primitive	failed	The exact curl (8.20) is nonzero.
Chronological future insertion	closed regular	Swapping the finite $j < k$ sums before conditioning gives (9.5); the high and fixed-low branches have slacks 1/14 and 17/42, and the terminal square stays unspent.
Weighted Carleson innovation–	closed regular	The stronger absolute bound (9.3) proves (9.1) for deterministic PSD heat on the one-shot class.

11 Devil’s-advocate review

- Objection: Heat-uniform boundedness of D^2B_R proves $\mathcal{K}_R$ by itself. UPHELD as false.** It controls coefficient amplitude, not multiplication on $\mathcal{C}^{-s}$ or the adapted innovation sum. The closure additionally needs the chronological identity (9.5), annular shell smoothing, prefix bounds, and one global Young allocation.
- Objection: Refunding the cubic retracts R-076. DISMISSED.** R-076 remains valid. Only its duplicate use inside the alternative full- L owner is forbidden.
- Objection: R-096 deletes global $T_G^>$. UPHELD as false.** It deletes the genuine large gap on each predictable single-insertion baseline. Equations (7.5)–(7.7) restore the future-feedback innovation in the global coordinate.
- Objection: R-101 value–gradient independence centres $d_j(Lc)$. UPHELD as false.** It centres the predictable local coefficient. The d_jc , product, and covariance-difference terms survive.
- Objection: Positivity of the Laplace–Wick formula gives positivity after taking its Taylor remainder. UPHELD as false.** The raw $\mathfrak{W}_2$ contribution in (4.6) is PSD, but the $\mathcal{R}_2^a$

and $\mathcal{R}_3^a$ remainders in (4.10), and hence their current crosses, are signless.

6. **Objection: The active lower bound applies to arbitrary c . UPHELD as false.** It requires $c \in \text{Ran } \Pi$, and no cross block may be discarded from a quadratic form.
7. **Objection: The kernel asymptotic disproves the paid theorem. DISMISSED.** It disproves global uniform inverse-Gram coercivity. It does not include Cameron–Martin payment or the complete signed owner.
8. **Objection: The finite-mode negative current is a Sector-A counterexample. DISMISSED.** It falsifies the naive bridge only; the square, payment, trace, forest, low term, and other rows are absent.
9. **Objection: The isotropic ridge settles every target heat. UPHELD as unsupported.** Formula (4.6) is all-covariance, but the ridge sign audit is one-dimensional and isotropic.
10. **Objection: Apply Young separately to every (j, k) pair. UPHELD as false.** That repeats the same Cameron–Martin and sextic budgets and violates one-use. Theorem 9.2 first swaps and sums every root.
11. **Objection: All-PSD heat formulas include floor removal. UPHELD as false.** The density floor remains fixed and positive.
12. **Objection: The low cross-Doob endpoint vanishes. UPHELD as false.** The term $\mathbb{E}\langle G_\ell, F_\ell \rangle$ in (7.2) is retained explicitly.
13. **Objection: Conditioning (9.4) drops a product covariance. DISMISSED.** Equation (9.5) conditions the complete product $\Delta_k \Psi^{(k)}$. Product and Jensen effects are contained in its exact future-insertion difference; no factorwise conditional product is asserted.
14. **Objection: A supremum over normalized Gaussian prefixes has a uniform moment. UPHELD as unnecessary and unsupported.** The proof uses only the geometrically weighted sequence norms (9.11), (9.13), whose sixth and fourteenth moments are summable by (9.14).
15. **Objection: Keep the R-086 $T_G^<$ payment after applying Theorem 9.2. UPHELD as duplicate ownership.** Theorem 9.2 estimates the complete current. Its active coordinate refunds that earlier current suballocation while leaving the R-086 theorem valid.
16. **Objection: Theorem 9.2 extends to arbitrary revisits. UPHELD as false.** Its pathwise physical-shell prefix bounds and annular last-insertion smoothing are specific to the regular no-revisit class. Overlapping progressive ranges require a separate complete-packet theorem.
17. **Objection: The fixed-low derivative in $c^{(k)}$ is a harmless additive constant. VALID with mitigation.** It can depend on the terminal control and therefore cannot be dropped from (9.12). Equation (9.10b) splits it explicitly; its monomial $X^{1/7} \bar{Y}^{19/42}$ has slack 17/42 and produces the second remainder in (9.16). The main 1/14 slack remains the limiting branch.

Executable adversarial audit. The primary certificate derives every production constant from the A1 manifest, symbolically computes the scalar owner values, first heat generator, and Cartan curl, and separately tests both isotropic Wick moments before their Laplace integration. A genuinely rank-deficient target covariance checks the determinant, mean, covariance, $\mathfrak{W}_1$, and $\mathfrak{W}_2$ tilting identities. A two-root finite filtration checks the actual cross-Doob brackets. A separate finite martingale array checks the $j < k$ swap and whole-product conditioning, a quartic polynomial checks deterministic heat composition, and exact rational ledgers check both chronological Hölder, interpolation, shell-decay, Young-slack,

and random-moment allocations. Active and kernel heat geometry is tested on deterministic Gaussian quadrature, with the asymptotic proof remaining analytic.

The independent certificate imports neither NumPy, SymPy, nor the primary program. Exact Fraction Taylor jets reproduce all scalar owner and bridge coefficients; exact rank-one Woodbury equations test singular heat; a separate bivariate jet derives the curl; finite probability enumeration checks the two-root Doob identity; Gaussian moment combinatorics derives the isotropic Wick coefficients; and an implementation-independent Fraction ledger reconstructs the chronological closure. External review is invited on the target-heat placement, physical-shell prefix hypotheses, R-086 refund, product-covariance owner, and sign convention in (7.6).

12 Executable verification

The primary executable passes 106 assertions and the standard-library independent executable passes 79 assertions. The integrated verifier reruns both, checks their exact common oracles, pins the upstream authorities and all package files, checks the proof note and PDF, and audits every public no- overclaim surface.

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_full_hessian_laplace_wick_future_feedback_boundary_verify.py
```

Result footer

Result ID: A13-CLASSII-FULL-HESSIAN-LAPLACE-WICK-FUTURE-
FEEDBACK-BOUNDARY

Ledger: R-102.

Title: Full-Hessian heat representation and regular future-insertion
rational-current closure.

Precise statement:

At fixed positive floor, the complete rational second-Hessian secant has a heat-uniform bounded tensor coefficient and an exact Laplace-Wick representation for every PSD target covariance. Full-L use requires refunding the separate base-cubic owner. The global current cross-Doob identity leaves an explicit future-feedback innovation residual. Swapping its finite $j < k$ sums before conditioning converts the whole residual into predictable last-insertion columns. Pathwise annular-shell estimates retain both the high-prefix and fixed-low derivative branches, with limiting Young slack $1/14$, uniformly in cutoff and deterministic PSD target heat. Thus regular K_R closes with the terminal square unspent. Exact scalar and Cartan fixtures still falsify shortcut factorizations; the Fourier row is a scoped diagnostic only.

Scope: Finite cutoff; fixed positive floor; local real-even coefficient; regular annular mutually orthogonal strict-past no-revisit controls for the owner and closure statements. All deterministic finite-dimensional PSD target covariances.

Dependencies:

A1; R-050; R-071; R-075; R-076; R-078; R-080; R-086;
R-092; R-096; R-099; R-100; R-101.

Evidence grade:

ANALYTIC, EXACT, EXECUTED, REGULAR-CLOSURE, BOUNDARY.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_full_hessian_laplace_wick_future_feedback_
boundary_verify.py
```

Expected output:

Manifest-pinned primary, standard-library independent,
integrated, and aggregate PASS; exit zero.

Explorations:

EXP-000250--EXP-000257.

Negative results:

NG-2026-07-28-A13-GLOBAL-T0-PREDICTABLE-CURRENT-BRIDGE;

NG-2026-07-28-A13-FULL-HESSIAN-CARTAN-CHAIN-PRIMITIVE.

Falsification gate:

Failure of full-L owner recombination, normalized Hessian formula, all-PSD Gaussian tilt or Laplace constants, isotropic Wick/ridge coefficients, active/kernel geometry, global cross-Doob identity, scalar bridge fixture, future-insertion identity, shell/Hoelder/Young ledger, Fourier diagnostic, Cartan curl, authority/hash/PDF checks, count contracts, or the stated T4 boundary.

Tier before / after:

T4 / T4; no promotion.

No-overclaim statement:

The weighted innovation-Carleson estimate and K_R lower form are proved only for the regular one-shot class. No full-frame posterior-covariance bracket, complete H_N, REG, progressive/revisit H_A, OVERLAP_src, Nelson, cutoff/floor removal, interacting measure, T5--T7 result, or Sector-A closure is proved.

Next required action:

Reassemble regular complete H_N and REG from the now-closed K_R row and the existing raw-Wick, low, linear, heat, trace, payment, q/r, and forest owners exactly once; then attack progression.